TE CASE STUDY REPORTS

Waste, E-Waste and Circular Economy

Ten case studies with one-page exam revision sheets

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1. Why Japan Recycles More Than Most Countries: Lessons for India
2. Switzerland's E-Waste Success Story
3. India's E-Waste Management Rules 2022
4. India's Extended Producer Responsibility (EPR): Success or Paper Compliance?
5. Bengaluru: India's E-Waste Capital?
6. Mumbai's E-Waste Ecosystem
7. Kerala's Clean Kerala Company
8. Pune's SWaCH Model
9. Dharavi Recycling Ecosystem
10. Dell's Circular Economy Strategy

Each case study: cover page, 2-page report, 1-page revision sheet.

Note on data: figures are rounded and reflect sources available up to mid-2026. Some are estimates or company-reported. Before submission, confirm key numbers against the latest editions of the cited sources and add access dates to the references.

1. Why Japan Recycles More Than Most Countries: Lessons for India

1. Executive Summary

Japan reaches very high recovery rates for packaging, metals and household appliances because it combines a layered legal framework, strict household segregation, shared producer-consumer-retailer duties and a cultural norm of avoiding waste (**mottainai**). Its overall municipal recycling rate is only about 20%, because most residual waste is incinerated with energy recovery; the accurate lesson is therefore "very little landfilling and high recovery in designated waste streams". India can adopt Japan's law design and segregation discipline, but must adapt them to its large informal sector and tight municipal budgets.

2. Background

After rapid post-war industrialisation, Japan faced pollution crises and scarce landfill space on a small, dense land area. The **Basic Act for Establishing a Sound Material-Cycle Society (2000)** set the waste hierarchy: reduce, reuse, recycle, thermal recovery, and only then disposal.

Product-specific laws followed: the **Containers and Packaging Recycling Law** (1995, in force 1997), the **Home Appliance Recycling Law** (2001), and laws for food, construction and end-of-life vehicles. The **Small Home Appliance Recycling Law** (2013) extended collection to phones, cameras and similar devices.

3. Strategy / Initiative

- **Legal ladder:** the Basic Act sets direction and targets; sector laws allocate duties to citizens, municipalities, retailers and manufacturers.
- **Household segregation:** most cities use roughly 5-15 waste categories with fixed collection days; Kamikatsu (Tokushima) asks residents to sort into about 45 categories and declared "zero waste" in 2003.
- **Cost sharing:** consumers pay a recycling fee when discarding TVs, air conditioners, refrigerators and washing machines; retailers take them back; manufacturers run the recycling plants.
- **Tracking:** a manifest system follows regulated appliances from retailer to recycler to deter illegal dumping.
- **Culture and education:** school programmes and community norms make sorting routine; the Tokyo 2020 medals were made entirely from donated small electronics.

4. Impact & Results

Indicator	Result / data	Source
Municipal solid waste generated	About 52 Mt (FY2000) reduced to about 40 Mt (FY2021)	[1]
Final disposal (landfill) of municipal waste	Roughly 10 Mt (FY2000) down to about 3.5 Mt (FY2021)	[1]
Overall municipal recycling rate	About 19-20% (rest mostly incinerated with energy recovery)	[1]
PET bottle recycling	Around 85% or higher (industry-reported)	[2]
Steel and aluminium cans	Around 90% (industry-reported)	[2]
Home appliances (4 legal categories)	Recovery rates reported above the legal minimums	[3]
Tokyo 2020 Olympic medals	About 5,000 medals from roughly 79,000 t of small e-devices, incl. about 6 million phones	[4]

Figures are rounded and taken from Japanese government and industry reports; check the latest edition before quoting exact values.

5. SWOT Analysis

STRENGTHS
- Clear, layered legislation with defined roles
- High citizen compliance and social norms
- Advanced sorting and incineration technology
- Consumer-pays fee for appliances

WEAKNESSES
- Overall recycling rate only about 20%
- Heavy reliance on incineration
- High collection labour and fee costs
- Some illegal dumping to avoid appliance fees

OPPORTUNITIES
- Shift from 3R to circular economy
- Urban mining of small electronics and batteries
- Technology transfer to developing countries (JICA)

THREATS
- Ageing population and labour shortage
- Export limits on plastic scrap since 2018
- Rising e-waste and EV battery volumes

6. Comparison with India

Aspect	Japan	India
Legal framework	Basic Act plus product laws with clear roles	SWM Rules 2016, E-Waste Rules 2022, Plastic Waste Rules; uneven enforcement

Aspect	Japan	India
Segregation	Mandatory multi-stream sorting, fixed days	Rules require segregation, but mixed collection is still common
Financing	Consumer end-of-life fee, producer-run plants	EPR certificates; municipal budgets and user fees are low
Informal sector	Minimal role	Central to collection and recycling
Disposal	Incineration dominant, little landfill	Large reliance on dumpsites and landfills
Behaviour	Strong compliance culture	Awareness and enforcement gaps

7. Recommendations

1. Enforce source segregation with reliable door-to-door collection, fixed timetables and graded fines plus rewards; scale up the Indore approach.
2. Introduce visible end-of-life or deposit fees on selected products to fund collection.
3. Integrate waste pickers through cooperatives (as in Pune) rather than displacing them.
4. Invest in material recovery facilities and material-specific recycling (PET, aluminium, e-waste).
5. Run school and community "zero-waste" pilots modelled on Kamikatsu.

8. Conclusion

Japan's performance is the product of law, infrastructure and behaviour working together, not culture alone. Its municipal recycling rate is modest, but its landfill reduction and stream-specific recovery are exceptional. India should copy the system design (clear duties, funded collection, enforced segregation) while building on its own strength: an informal workforce that already recovers large volumes of material.

9. References

[1] Ministry of the Environment, Japan. Annual Report on the Environment, the Sound Material-Cycle Society and Biodiversity (latest edition).
[2] PET Bottle Recycling Council (Japan) and Japan Aluminium Can Recycling Association annual data.
[3] METI / Ministry of the Environment, Japan. Home Appliance Recycling Law: annual implementation results.
[4] Tokyo 2020 Organising Committee. Tokyo 2020 Medal Project: Towards an Innovative Future for All (final report).
[5] MoEFCC, India. Solid Waste Management Rules, 2016; E-Waste (Management) Rules, 2022.

Exam Revision Sheet | Case Study 1: Japan Recycling

30-SECOND ANSWER

Japan recycles well because of **strong laws + segregation + shared costs + culture**. Overall municipal recycling is only about 20% (most residue is burned), but landfill fell by about two-thirds since 2000 and appliances, PET and cans are recovered at very high rates. India should copy the system, not just the culture.

KEY FACTS

- Basic Act for a Sound Material-Cycle Society: 2000
- Packaging law 1995 (in force 1997); Home Appliance law 2001; Small Appliance law 2013
- Hierarchy: reduce, reuse, recycle, thermal recovery, disposal
- Consumer pays fee at disposal for 4 appliance types
- Kamikatsu: about 45 categories, zero-waste declaration 2003
- Tokyo 2020 medals from donated e-devices

KEY TERMS

Mottainai: regret at waste; cultural value

3R: reduce, reuse, recycle

Sound Material-Cycle Society: Japan's circular-economy vision

Manifest system: paper/e-tracking of waste to prevent illegal dumping

Thermal recovery: incineration with energy recovery

NUMBERS & DATES TO REMEMBER

2000: Basic Act | 2001: Appliance law | 2013: Small appliance law | MSW recycling about 20% | Landfill: about 10 Mt to about 3.5 Mt | About 5,000 medals

STRENGTHS / WHAT WORKED

- Law + culture + tech together
- Very low landfilling
- Clear producer/consumer roles
- Stream-specific high recovery

LIMITS / CRITICISMS

- Overall recycling only about 20%
- Incineration dependence
- High cost, labour shortage
- Illegal dumping of appliances

LIKELY EXAM QUESTIONS

1. Explain why Japan is called a leader in recycling. Is the claim fully accurate?
2. Describe the Home Appliance Recycling Law and who pays for what.
3. List three lessons for India and one barrier for each.
4. Compare Japan and India using four parameters.

Memory hook: "BASIC-SORT-PAY-TRACK": Basic Act, Sort at source, Pay at disposal, Track with manifests.

2. Switzerland's E-Waste Success Story

1. Executive Summary

Switzerland has one of the world's highest e-waste collection rates because it made return free for the consumer, made take-back compulsory for retailers and manufacturers, and financed everything through a visible **Advance Recycling Fee (ARF)** paid at the time of purchase. Producer-run organisations (SENS, Swico, SLRS) run the system under a 1998 ordinance, and the fee model has lasted for decades with little public resistance. It shows that an industry-managed, fee-funded EPR system can work when rules are simple and enforcement is credible.

2. Background

Switzerland has a high income, high electronics consumption (over 20 kg of e-waste generated per person per year, among the highest in Europe) and very limited landfill space. Recycling of fridges began voluntarily in the early 1990s, and industry bodies introduced fee-based systems before the law required them.

The **Ordinance on the Return, Taking Back and Disposal of Electrical and Electronic Equipment (ORDEO / VREG)** came into force on **1 July 1998**, converting successful voluntary schemes into a national legal duty.

3. Strategy / Initiative

- **Three duties:** consumers must return old devices; retailers, importers and manufacturers must take them back free of charge; producers must ensure environmentally sound disposal.
- **Advance Recycling Fee (ARF):** a small fee added to the price of each new device and shown on the receipt, pooled to pay for transport and recycling. No cost at the time of disposal removes the incentive to dump.
- **Producer-run organisations:** SENS eRecycling (household appliances, tools), Swico (IT, office, telecom, consumer electronics) and SLRS (lamps and luminaires) operate collection and audited recyclers.
- **Dense free take-back network:** thousands of retail and municipal collection points; drop-off is possible anywhere devices are sold.
- **Quality control:** recyclers are certified and audited; data on flows is reported to the federal environment office (FOEN).
- **Knowledge export:** Empa-led programmes (funded by the Swiss government) have supported e-waste capacity building in India and other countries.

4. Impact & Results

Indicator	Result / data	Source
Legal basis	ORDEO in force since 1 July 1998	[1]
Collection per person	Roughly 15 kg per person per year, among the world's highest	[2]
Total collected and recycled	More than 100,000 t per year (SENS + Swico + SLRS)	[2][3]
Material recovery	Typically above 90% by weight for most appliance types	[3]
Global comparison	World: 22.3% of e-waste formally collected and recycled (2022); Europe: about 43%	[4]
Funding model	Visible ARF paid at purchase; no charge on return	[2]

Swiss collection figures vary by year and definition; treat values as approximate and confirm against the latest SENS, Swico and FOEN reports.

5. SWOT Analysis

STRENGTHS

- Simple, uniform rule: return free, pay at purchase
- Trusted industry organisations
- Very dense collection network
- High consumer awareness

WEAKNESSES

- Depends on high incomes and small market
- ARF must be set correctly for new products
- Free-riders from online sellers abroad
- Some valuable devices stored at home

OPPORTUNITIES

- Better recovery of critical raw materials
- Repair and reuse expansion
- Extend to batteries, solar panels, e-bikes

THREATS

- Cross-border online sales avoiding the fee
- Fire risk from lithium batteries in collections
- Falling material prices reducing recycler revenue

6. Comparison with India

Aspect	Switzerland	India
Legal basis	ORDEO 1998; clear consumer and producer duties	E-Waste Rules 2022 with EPR targets for producers

Aspect	Switzerland	India
Funding	Visible ARF at purchase	EPR certificates traded between producers and recyclers
Consumer role	Legally bound to return; free drop-off	Voluntary; most devices sold to kabadiwalas for cash
Collection network	Thousands of retail and municipal points	Limited authorised collection points and reverse logistics
Informal sector	Negligible	Handles the majority of e-waste
Enforcement	Audited, transparent flows	Traceability and audit capacity still developing

7. Recommendations

1. Pilot a small, visible recycling fee at point of sale in selected product categories, with drop-off free.
2. Require retailers and service centres to accept old devices when new ones are sold.
3. Formalise kabadiwalas as authorised collectors paid per kg, instead of banning them.
4. Publish plant-level flow data so recyclers and PROs can be audited by the public.
5. Set up certified dismantling training modelled on Swiss-Empa programmes.

8. Conclusion

Switzerland shows that convenience, transparency and shared responsibility matter more than penalties. Its ARF model is not directly transferable to India because incomes and market structure differ, but its principles (free return, funded collection, audited recyclers) fit within India's EPR framework, especially if the informal sector is brought into the formal chain.

9. References

[1] Swiss Federal Council. Ordinance on the Return, Taking Back and Disposal of Electrical and Electronic Equipment (ORDEO/VREG), 1998.
[2] Swiss Federal Office for the Environment (FOEN). Waste: Electrical and electronic equipment (web pages and waste statistics).
[3] SENS eRecycling, Swico Recycling and SLRS. Annual reports (latest editions).
[4] Forti, V. et al. The Global E-waste Monitor 2024. UNITAR and ITU, 2024.
[5] Empa (Swiss Federal Laboratories). Sustainable Recycling Industries programme documents.

Exam Revision Sheet | Case Study 2: Switzerland E-Waste

30-SECOND ANSWER

Switzerland: **free return + mandatory take-back + visible Advance Recycling Fee**, run by producer organisations (SENS, Swico, SLRS) under a 1998 ordinance. Result: about 15 kg per person collected each year and 90%+ material recovery. India can borrow the principles, not the exact fee model.

KEY FACTS

- ORDEO in force 1 July 1998
- SENS: appliances; Swico: IT/electronics; SLRS: lamps
- ARF added to price and shown on receipt
- Retailers must take back free
- Recyclers certified and audited
- Empa supports e-waste capacity building in India

KEY TERMS

ARF: Advance Recycling Fee paid at purchase

EPR: producer responsibility for end-of-life products

PRO: producer responsibility organisation

ORDEO/VREG: Swiss e-waste ordinance

NUMBERS & DATES TO REMEMBER

1998: ORDEO | About 15 kg per person collected | >100,000 t per year | >90% material recovery | World collection 22.3% vs Europe about 43%

STRENGTHS / WHAT WORKED

- Simple, transparent model
- Convenient free drop-off
- Industry-managed, audited
- Long-running with public trust

LIMITS / CRITICISMS

- Needs high income and small market
- Online sales from abroad
- Battery fire risk
- Fee must be updated

LIKELY EXAM QUESTIONS

1. Explain the ARF model and why consumers accept it.
2. Compare Swiss producer organisations with India's PROs and EPR certificates.
3. What can India adopt from Switzerland and what cannot be copied?
4. Explain the three duties under ORDEO.

Memory hook: "Return FREE, Pay EARLY": you pay the fee when you buy, not when you dispose.

3. India's E-Waste Management Rules 2022

1. Executive Summary

The **E-Waste (Management) Rules, 2022**, notified on 2 November 2022 and in force from **1 April 2023**, replaced the 2016 rules and moved India from a broad EPR obligation to a **certificate-based, portal-driven system**. They widen coverage from 21 to 106 product types, require registration of all manufacturers, producers, refurbishers and recyclers, set annual recycling targets that rise from 60% to 80%, and ban hazardous substances above limits. The main question is enforcement: whether certificates correspond to real, traceable recycling.

2. Background

India generated about 4.1 million tonnes of e-waste in 2022 according to the Global E-waste Monitor 2024, and India's own CPCB data record a rise from about 0.7 Mt in 2017-18 to about 1.6 Mt in 2021-22 (recorded volumes). A large share of collection and dismantling is informal, where unsafe methods release lead, mercury and brominated compounds.

Regulation evolved in steps: **2011 Rules** (in force 2012) introduced EPR; **2016 Rules** added collection targets, Producer Responsibility Organisations and a deposit-refund option; **2022 Rules** introduced EPR certificates, centralised registration and financial penalties.

3. Strategy / Initiative

- **Scope:** Schedule I covers 106 electrical and electronic items (up from 21), including solar PV modules and many consumer and IT products.
- **Registration:** manufacturers, producers, refurbishers and recyclers must register on the CPCB portal; unregistered operations are not allowed.
- **EPR targets:** producers must meet annual targets (60% of past sales in the first years, rising to 80%) by purchasing EPR certificates generated by registered recyclers or refurbishers. Certificates are created only when e-waste is actually processed or refurbished.
- **Floor and ceiling prices:** the pricing of certificates is regulated by CPCB to prevent under-priced trading.
- **Reduction of hazardous substances (RoHS):** limits of 0.1% for lead, mercury, chromium VI, PBB and PBDE, and 0.01% for cadmium, in new products.
- **Refurbishing and reuse:** refurbishers become part of the system and can generate certificates, extending product life.
- **Penalties:** environmental compensation for non-compliance, and audit and verification by CPCB / SPCBs.

4. Impact & Results

Indicator	Result / data	Source
Rules in force	1 April 2023 (notified 2 Nov 2022)	[1]
Items covered	106 (was 21 under 2016 Rules)	[1]
Recorded e-waste generation	About 0.71 Mt (2017-18), 1.01 Mt (2019-20), 1.60 Mt (2021-22)	[2]
Estimated total (GEM 2024)	About 4.1 Mt generated in India, 2022	[3]
EPR target	60% initially, rising to 80% of the base sales	[1]
Formal share	Formal collection remains a minority; informal sector dominates	[2][3]

CPCB figures depend on reporting by registered entities and differ from modelled estimates such as GEM 2024. Confirm the latest CPCB annual report before quoting.

5. SWOT Analysis

STRENGTHS
- Wider coverage (106 items)
- Digital registration and certificates
- Includes refurbishers and RoHS
- Regulated certificate pricing

WEAKNESSES
- Informal sector not well integrated
- Verification of actual recycling is hard
- Low consumer awareness
- Limited reverse logistics for households

OPPORTUNITIES
- Growth of formal recycling industry and urban mining
- Digital traceability using QR / blockchain
- Refurbishment and repair economy

THREATS
- Fraudulent or inflated certificates
- Rapid growth of e-waste and lithium batteries
- Recyclers cannot meet capacity or utilisation targets

6. Comparison with India

Aspect	E-Waste Rules 2022	India
Scope	106 items, includes solar PV	EU WEEE covers all EEE in 6 categories
Producer duty	Certificate-based EPR with targets	EU/Swiss: physical collection targets with take-back
Consumer role	No legal duty; voluntary	Switzerland: legal duty to return
Informal sector	Not formally integrated	Practically absent in Switzerland/EU
Data and audits	Portal-based, improving	Mature audited systems

7. Recommendations

1. Link EPR certificates to verifiable collection weights with QR tags and audit trails.
2. Create authorised collector status and payment for informal collectors.
3. Require retailers and service centres to run take-back points.
4. Run awareness campaigns on home e-waste, data security and battery safety.
5. Publish state-wise audit and penalty data every year.

8. Conclusion

The 2022 Rules are a significant modernisation, with wider scope, digital registration and regulated certificates. Their success will depend less on the text and more on verification, informal-sector integration and consumer participation.

9. References

[1] MoEFCC. E-Waste (Management) Rules, 2022. Gazette notification, 2 November 2022.
[2] CPCB. Annual Reports on Implementation of E-Waste Rules (various years).
[3] Forti, V. et al. The Global E-waste Monitor 2024. UNITAR and ITU, 2024.
[4] CPCB. Guidelines for Implementation of EPR under E-Waste Rules; EPR portal documentation.

Exam Revision Sheet | Case Study 3: E-Waste Rules 2022

30-SECOND ANSWER

E-Waste Rules 2022 (in force 1 April 2023) replaced 2016 rules: **106 items, mandatory registration, EPR certificates, targets 60% rising to 80%, RoHS limits, refurbishers included.** Key challenge: verifying real recycling and integrating the informal sector.

KEY FACTS

- Notified 2 Nov 2022; in force 1 Apr 2023
- Coverage 21 items (2016) to 106 items
- CPCB portal for registration and certificates
- Producers buy EPR certificates from recyclers/refurbishers
- RoHS: Pb, Hg, Cr VI, PBB, PBDE at 0.1%; Cd at 0.01%
- Environmental compensation for violations

KEY TERMS

EPR certificate: proof of processed e-waste, traded

Producer: brand owner / importer / seller under brand

Refurbisher: restores used EEE for reuse

RoHS: restriction of hazardous substances

Floor price: minimum certificate price

NUMBERS & DATES TO REMEMBER

2011 (first EPR) | 2016 rules | 2022 rules | 106 items | 60% to 80% targets | India about 4.1 Mt (2022)

STRENGTHS / WHAT WORKED

- Broader scope
- Digital tracking
- Regulated certificate prices
- Reuse recognised

LIMITS / CRITICISMS

- Informal sector excluded
- Paper compliance risk
- Weak reverse logistics
- Limited awareness

LIKELY EXAM QUESTIONS

1. Explain key features of E-Waste Rules 2022 and how they differ from 2016.
2. How does the EPR certificate system work?
3. Discuss challenges in implementation.
4. What is RoHS and why is it included?

Memory hook: "106-60/80-Portal-RoHS": items, targets, portal, hazardous limits.

4. India's Extended Producer Responsibility (EPR): Success or Paper Compliance?

1. Executive Summary

EPR makes producers responsible for the end-of-life of what they sell. India introduced e-waste EPR in 2011-12 and rebuilt it in 2022 around tradeable **EPR certificates**. The design is sound and registrations, recycling capacity and recorded volumes have grown. But the system has weaknesses: certificates can be produced without genuinely improving collection, informal recyclers still handle most e-waste, and verification is limited. This case study concludes that EPR in India is **partially successful in structure but exposed to paper compliance in practice**.

2. Background

EPR moves the cost of collection and recycling from municipalities and society to producers. In India it appears in E-Waste Rules (2011, 2016, 2022), Plastic Waste Rules, Battery Waste Rules (2022) and Tyre and other categories.

Early e-waste EPR relied on producer plans and often produced weak results: most producers reported small collection volumes and few consumers used take-back. The 2022 Rules replaced plans with certificate targets to make outcomes measurable.

3. Strategy / Initiative

- **Registration** of producers, recyclers and refurbishers on a central portal.
- **Annual EPR targets** as a percentage of past sales (rising to 80%).
- **Certificates** generated only when registered recyclers process e-waste; producers buy them to meet targets.
- **Producer Responsibility Organisations (PROs)** manage compliance for many brands and organise collection.
- **Regulated prices** through floor and ceiling limits, and environmental compensation for shortfalls.
- **Audit and verification** by CPCB and State Pollution Control Boards.

4. Impact & Results

Indicator	Result / data	Source
Start of e-waste EPR	2011 rules (in force 2012)	[1]
Current framework	2022 Rules: certificates, portal, targets	[1]
Recorded processed share	Roughly one-third or less of recorded generation (CPCB reports)	[2]
Recycling capacity	Authorised capacity exceeds volumes actually processed; utilisation is low	[2]
Informal sector share	Commonly estimated at a large majority of e-waste handling	[3]
Integrity risk	Concerns about fake or inflated certificates (reported mainly in plastic EPR)	[4]

Qualitative points on integrity are based on reported audits and press coverage, not on a single official statistic; verify before citing in an exam or viva.

5. SWOT Analysis

STRENGTHS
- Clear legal mandate and targets
- Portal creates a data trail
- Growing formal recycling industry
- Producers now financially involved

WEAKNESSES
- Targets are on paper (certificates) rather than physical collection
- Weak audit capacity
- Households rarely engage
- Informal sector excluded

OPPORTUNITIES
- Traceability tools (QR, blockchain, weighbridge data)
- Take-back through retail and service centres
- Design-for-recycling incentives

THREATS
- Certificate fraud and double counting
- Recyclers relying on cheap informal inputs
- Producer non-compliance and weak penalties

6. Comparison with India

Aspect	Ideal EPR (EU / Switzerland)	India
Design	Target-based with tradeable certificates	EU: physical collection targets, mandatory take-back
Funding link	Producer pays via certificate purchase	Fee at sale (Switzerland) or PRO fees (EU)
Verification	Portal + selected audits	Regular independent audits, mature
Informal sector	Outside the chain	Marginal

Aspect	Ideal EPR (EU / Switzerland)	India
Consumer participation	Low	High and often legally required

7. Recommendations

1. Tie certificates to physical evidence (weighbridge, video, QR).
2. Increase surprise audits and publish penalties.
3. Add mandatory take-back at retail and service centres.
4. Pay informal collectors as registered sub-contractors.
5. Make eco-design (repairable, recyclable products) a part of producer fees.

8. Conclusion

India's EPR is a well-designed framework whose effectiveness depends on data integrity. Without stronger verification and consumer access to take-back, it risks becoming a market in certificates rather than a system that actually recovers more e-waste.

9. References

[1] MoEFCC. E-Waste (Management) Rules 2011, 2016 and 2022.
[2] CPCB. Annual reports on e-waste management and EPR portal data.
[3] Forti, V. et al. The Global E-waste Monitor 2024. UNITAR and ITU, 2024.
[4] Press and CPCB communications on EPR audits and certificate irregularities (2024-25); cite specific articles used.

Exam Revision Sheet | Case Study 4: EPR: Success or Paper?

30-SECOND ANSWER

India's EPR (2011, rebuilt 2022) makes producers pay through **tradeable certificates**. It builds a formal industry and data trail but is vulnerable to **paper compliance**: certificates without real collection, low audit capacity, informal sector left out, low consumer take-back.

KEY FACTS

- EPR: producers responsible for end-of-life
- Certificates only from registered recyclers/refurbishers
- PROs handle compliance for many producers
- Floor/ceiling prices regulate certificates
- Environmental compensation for non-compliance

KEY TERMS

EPR: extended producer responsibility

PRO: producer responsibility organisation

Paper compliance: meeting targets on paper, not in reality

Traceability: tracking material from source to recycler

NUMBERS & DATES TO REMEMBER

2011/12: EPR starts | 2022: certificate system | Targets 60% to 80% | Low utilisation of capacity

STRENGTHS / WHAT WORKED

- Producer financial responsibility
- Digital portal
- Formal capacity grown
- Clear targets

LIMITS / CRITICISMS

- Fraud risk
- Weak audits
- Informal sector excluded
- Low consumer participation

LIKELY EXAM QUESTIONS

1. Is EPR in India a success or paper compliance? Argue with examples.
2. Explain how EPR certificates work.
3. Suggest reforms to strengthen EPR.
4. Compare Indian EPR with Switzerland / EU.

Memory hook: "Certificate is not collection": always ask what physically happened to the e-waste.

5. Bengaluru: India's E-Waste Capital?

1. Executive Summary

Bengaluru, India's IT and start-up hub, is a major e-waste generator and one of the country's most important recycling centres. Whether it is India's "capital" depends on the measure: older industry estimates rank Mumbai and Delhi-NCR higher by volume, while Bengaluru leads in corporate (B2B) IT e-waste, organised recyclers and informal repair-and-resale markets. The city shows both the opportunity of a formal, technology-driven recycling industry and the difficulty of controlling the informal chain.

2. Background

Bengaluru hosts a large concentration of IT/BPM firms, start-ups, data centres, electronics retail and service centres, so the city generates large volumes of discarded computers, servers, phones and batteries. ASSOCHAM-NEC (2016) estimated about 92,000 t of e-waste per year for Bengaluru, ranking it third among Indian cities after Mumbai and Delhi-NCR. These are dated estimates and later data are less city-specific.

Karnataka is consistently among the top states in CPCB e-waste reporting, and Bengaluru is home to some of India's earliest authorised recyclers, such as **E-Parisaraa** (Dobbaspet), started around 2006.

3. Strategy / Initiative

- **Authorised recycling:** KSPCB-authorised dismantlers and recyclers process corporate and institutional e-waste with EPR documentation.
- **Corporate take-back:** IT companies and OEMs send end-of-life IT assets to certified recyclers, often with data destruction.
- **Local collection drives:** civic bodies, RWAs, NGOs and companies run periodic drop-off events; some retailers run take-back.
- **Waste-picker networks:** organisations such as Hasiru Dala and Saahas Zero Waste work on dry waste collection and awareness, and link some informal workers with authorised channels.
- **Informal markets:** repair and resale clusters (for example around S.P. Road and Kalasipalya) extend product life but also feed informal dismantling.

4. Impact & Results

Indicator	Result / data	Source
City estimate (ASSOCHAM-NEC, 2016)	About 92,000 t per year (3rd after Mumbai, Delhi-NCR)	[1]
National total (GEM 2024)	About 4.1 Mt generated, India, 2022	[2]
Corporate e-waste	Large B2B share due to IT/BPM concentration	[1]
Authorised recyclers	Multiple KSPCB-authorised units in and around the city	[3]
Informal handling	Substantial share still handled by informal dismantlers	[1][2]

City-level e-waste data in India are mostly estimates. Use the numbers to show scale, not exact ranking, and check for a more recent study.

5. SWOT Analysis

STRENGTHS

- Strong B2B e-waste flow to authorised recyclers
- Early formal recycling industry
- Tech talent and start-ups for recycling innovation
- Active NGOs and waste-picker groups

WEAKNESSES

- Weak household take-back
- Large informal dismantling market
- Data on city flows are patchy
- Air and soil pollution near informal sites

OPPORTUNITIES

- Traceability apps and digital take-back
- Refurbishment and repair-as-a-service
- Green jobs and skills for informal workers

THREATS

- Rapid device turnover and lithium batteries
- Leakage of B2B e-waste to informal buyers
- Land pressure pushing recyclers to city edges

6. Comparison with India

Aspect	Bengaluru	India
Scale	Top-3 city generator in older estimates	Mumbai and Delhi-NCR rank comparable or higher
Formal industry	Early, organised recyclers; B2B strength	Formal capacity concentrated in a few states
Household channel	Weak, event-based	Same nationwide
Informal sector	Large repair and resale clusters	Dominant across cities
Enforcement	KSPCB inspections, uneven	Uneven across states

7. Recommendations

1. Establish permanent household drop-off points at malls, ward offices and service centres.
2. Require corporates to submit disposal certificates, data-wipe proof and recycler audits.
3. Register informal dismantlers as authorised collectors and give safety training and PPE.
4. Publish city-level e-waste flow data annually.
5. Support refurbishing and repair start-ups with incentives.

8. Conclusion

Bengaluru may not be India's largest e-waste generator on every measure, but it is a leading test bed for formal e-waste recycling. Making it a model city requires controlling leakage from corporate flows and integrating the informal repair economy into the authorised system.

9. References

[1] ASSOCHAM and NEC. Electronic Waste Management in India (study, 2016).

[2] Forti, V. et al. The Global E-waste Monitor 2024. UNITAR and ITU, 2024.

[3] Karnataka State Pollution Control Board (KSPCB). List of authorised e-waste dismantlers/recyclers.

[4] E-Parisaraa Pvt. Ltd. company information; Hasiru Dala and Saahas Zero Waste organisational reports.

Exam Revision Sheet | Case Study 5: Bengaluru E-Waste

30-SECOND ANSWER

Bengaluru is a **major e-waste hub** (IT/BPM, startups; about 92,000 t/year in 2016 estimate, third after Mumbai and Delhi), with early formal recyclers (E-Parisaraa) but a large informal repair and dismantling economy. "Capital" is debatable; it is strongest in **B2B e-waste and recycling capacity**.

KEY FACTS

- Estimate: about 92,000 t/yr (ASSOCHAM-NEC 2016)
- E-Parisaraa: early authorised recycler (around 2006)
- KSPCB authorises recyclers
- Informal clusters: S.P. Road, Kalasipalya
- NGOs: Hasiru Dala, Saahas Zero Waste

KEY TERMS

B2B e-waste: discarded by businesses (large, traceable)

Leakage: authorised waste going to informal buyers

Urban mining: recovering metals from e-waste

NUMBERS & DATES TO REMEMBER

About 92,000 t (2016) | Rank 3 city | India about 4.1 Mt (2022)

STRENGTHS / WHAT WORKED

- Formal recyclers
- B2B strength
- Tech innovation
- Active NGOs

LIMITS / CRITICISMS

- Household take-back weak
- Informal sector
- Old data
- Leakage risk

LIKELY EXAM QUESTIONS

1. Is Bengaluru India's e-waste capital? Discuss with data.
2. Explain the role of formal and informal sectors in Bengaluru.
3. Suggest a plan to improve e-waste management in Bengaluru.

Memory hook: "B-B-B": Bengaluru = B2B waste, Big IT, Both formal and informal.

6. Mumbai's E-Waste Ecosystem

1. Executive Summary

Mumbai's e-waste ecosystem is a dense chain of consumers, retailers, kabadiwalas, informal dismantling clusters and a small number of authorised recyclers, embedded in a state (Maharashtra) that is consistently a top e-waste generator. It has large flows from households, offices and electronics markets, and a mixture of best-practice recyclers and hazardous informal recovery. Improving Mumbai's system means connecting the existing informal network to authorised recyclers rather than building a parallel system.

2. Background

Mumbai is India's financial capital with high device ownership, a large corporate sector and big electronics markets such as Lamington Road. ASSOCHAM-NEC (2016) estimated Mumbai as India's largest city generator at about 1.2 lakh tonnes per year, and Maharashtra has consistently ranked among the top states in CPCB and industry studies.

Scarce land, high property values and a large informal economy shape where and how dismantling occurs: clusters in Govandi-Mankhurd, Dharavi and Bhiwandi-Thane handle a large share of discarded electronics.

3. Strategy / Initiative

- **Generation and collection:** households sell old devices to kabadiwalas; offices use AMC vendors or recyclers; retailers provide exchange offers.
- **Informal aggregation and dismantling:** manual breaking of PCBs, cables and plastics, and acid-based metal recovery in small units.
- **Authorised recyclers:** MPCB-authorised units such as Eco Recycling Ltd (Ecoreco) provide certified dismantling and downstream processing.
- **Municipal and civil initiatives:** BMC and NGOs run periodic e-waste collection days and awareness campaigns; producers and PROs organise brand take-back.
- **Regulation:** MPCB authorisation and monitoring under the E-Waste Rules 2022.

4. Impact & Results

Indicator	Result / data	Source
City estimate (ASSOCHAM-NEC, 2016)	About 1.2 lakh t/yr, largest among Indian cities	[1]
State rank	Maharashtra consistently among top e-waste-generating states	[2]
National estimate	India about 4.1 Mt (2022)	[3]
Informal clusters	Govandi-Mankhurd, Dharavi, Bhiwandi-Thane	[1]
Formal recyclers	Several MPCB-authorised units in the Mumbai region	[4]
Health/environment	Documented risks from open burning and acid leaching	[1][3]

City data are dated estimates. Cite the year and source whenever you use them and check for newer MPCB/CPCB numbers.

5. SWOT Analysis

STRENGTHS
- Large, concentrated flows (efficient logistics)
- Established informal network with skills
- Presence of authorised recyclers
- Active civil society

WEAKNESSES
- Informal dismantling is hazardous
- Space constraints for new plants
- Low household awareness
- Fragmented data

OPPORTUNITIES
- Partnerships between recyclers and kabadiwalas
- Digital take-back platforms
- Green skilling in Dharavi and Govandi

THREATS
- Land pressure and evictions
- Illegal dumping and burning
- Growth of lithium battery waste

6. Comparison with India

Aspect	Mumbai	India
Scale	Largest city generator (2016 estimate)	Comparable to Delhi-NCR and Bengaluru
Chain	Kabadiwala to cluster to trader; small formal share	Same national pattern
Land and cost	Very high land costs limit plants	Less severe in smaller cities
Regulation	MPCB authorisation, uneven enforcement	Uneven
Best practice	Ecoreco-type formal processing	Switzerland: universal formal chain

7. Recommendations

1. Give kabadiwalas ID cards and a per-kg price from authorised recyclers.
2. Create ward-level drop-off points and periodic collection days with BMC.
3. Provide safe common facilities for dismantling in industrial zones.
4. Mandate corporate disposal audits and consumer awareness through retailers.
5. Monitor air, water and soil around informal clusters and rehabilitate contaminated sites.

8. Conclusion

Mumbai illustrates how a large city can have all the elements of a formal system yet still lose most e-waste to informal channels. Better prices, certification and safe facilities for informal actors can raise formal recovery without destroying livelihoods.

9. References

[1] ASSOCHAM and NEC. Electronic Waste Management in India (study, 2016).

[2] CPCB. Annual reports on e-waste management (various years).

[3] Forti, V. et al. The Global E-waste Monitor 2024. UNITAR and ITU, 2024.

[4] Maharashtra Pollution Control Board (MPCB). List of authorised e-waste recyclers; Ecoreco company information.

Exam Revision Sheet | Case Study 6: Mumbai E-Waste

30-SECOND ANSWER

Mumbai: **largest city generator in the 2016 estimate (about 1.2 lakh t/yr)**, in a top state (Maharashtra). Chain: households/offices to kabadiwalas to informal clusters (Govandi-Mankhurd, Dharavi, Bhiwandi) with a few authorised recyclers (Ecoreco). Solution: **link informal collectors to authorised recyclers.**

KEY FACTS

- Electronics market: Lamington Road
- Clusters: Govandi-Mankhurd, Dharavi, Bhiwandi-Thane
- MPCB authorises recyclers
- Ecoreco: Mumbai-based authorised recycler
- BMC/NGO collection drives

KEY TERMS

Kabadiwala: informal scrap dealer

AMC: annual maintenance contract vendor

Acid leaching: hazardous metal recovery method

NUMBERS & DATES TO REMEMBER

About 1.2 lakh t (2016) | India about 4.1 Mt (2022)

STRENGTHS / WHAT WORKED

- Large flows
- Skilled informal network
- Formal recyclers exist

LIMITS / CRITICISMS

- Hazardous informal methods
- Land constraints
- Poor data

LIKELY EXAM QUESTIONS

1. Describe the flow of e-waste in Mumbai from consumer to recovery.
2. What are the risks of informal recycling and how can they be reduced?
3. Suggest measures to formalise Mumbai's e-waste chain.

Memory hook: "Collect-Cluster-Certify": kabadiwala collects, cluster dismantles, recycler certifies.

7. Kerala's Clean Kerala Company

1. Executive Summary

The **Clean Kerala Company Limited (CKCL)** is a Government of Kerala company under the Local Self Government Department that manages non-biodegradable and hazardous waste, including e-waste, by linking local bodies, community collection and authorised recyclers. Working with Haritha Karma Sena (HKS), the women-led green army that collects dry waste door to door for a user fee, it demonstrates a public-sector, community-based model. Its strengths are decentralised collection and clear institutional roles; its challenges are user-fee collection, market prices for recyclables and scale.

2. Background

Kerala has high population density, high literacy and consumption, heavy rainfall and scarce land for landfills, and it faced strong public protests against dumpsites. The state's Suchitwa Mission and Haritha Keralam Mission led to a decentralised approach in which local self-government institutions (LSGIs) manage waste with community support.

CKCL was set up to be the state-level agency handling the non-biodegradable stream: buying back, aggregating and sending recyclables, and organising e-waste and hazardous-waste collection.

3. Strategy / Initiative

- **Haritha Karma Sena (from about 2017):** local women's groups collect non-biodegradable waste door to door for a monthly user fee and hand it to Material Collection Facilities (MCFs).
- **Aggregation:** MCFs and Resource Recovery Facilities sort and bale material; CKCL arranges transport and sale or recycling.
- **E-waste and hazardous waste drives:** periodic collections from households, government offices and institutions, sent to authorised recyclers.
- **Plastic processing:** shredded plastic is used in road tarring and cement kilns under regulations.
- **Financing and support:** LSGD funding, user fees, revenue from recyclables and World Bank-supported programmes (KSWMP).
- **Monitoring:** digital tracking of collection coverage and material flows.

4. Impact & Results

Indicator	Result / data	Source
Institutional model	State company + LSGIs + HKS members	[1]
Haritha Karma Sena workforce	Tens of thousands of members, mostly women	[1][2]
Coverage	Door-to-door collection in most urban and many rural local bodies	[1]
E-waste	Regular collection drives and delivery to authorised recyclers	[1]
Plastic reuse	Shredded plastic used in road construction	[2]

Numbers for HKS membership, households covered and tonnage change yearly; obtain the latest values from CKCL or Suchitwa Mission reports before writing exact figures.

5. SWOT Analysis

STRENGTHS
- Strong institutional mandate and community links
- Women-led job creation
- Decentralised, low-transport model
- High public awareness in Kerala

WEAKNESSES
- User fee non-payment in some areas
- Dependence on volatile recyclable prices
- Limited handling of low-value plastics
- Capacity gaps in MCFs

OPPORTUNITIES
- Integrating e-waste and battery collection
- Certified recycling partnerships
- Digital payment and tracking

THREATS
- Recyclable market crashes
- Political and funding changes
- Rising e-waste and multilayer plastic

6. Comparison with India

Aspect	Kerala (CKCL)	India
Institution	State company with LSGI/HKS network	Municipal corporations, varying models
Workforce	Women-led green army, user fee	Informal pickers or contractors
Coverage	Widespread door-to-door dry waste	Patchy; mixed collection in many cities
Financing	User fee + state funds + material sales	ULB funds, low user fees

Aspect	Kerala (CKCL)	India
E-waste	Organised drives with authorised recyclers	Limited; informal chains dominate

7. Recommendations

1. Replicate HKS-style group enterprises with a legal user-fee structure.
2. Ensure fair wages, ID and social security for members.
3. Create standard MCF design with e-waste and battery segregation cells.
4. Link CKCL-type aggregators with PROs for EPR funding.
5. Use digital tracking to prove weights and reduce leakage.

8. Conclusion

The Clean Kerala model shows that a public company can coordinate community collectors, local bodies and recyclers. To scale, it needs stable financing, better handling of low-value waste and integration with EPR systems.

9. References

[1] Clean Kerala Company Limited (CKCL). Official website and annual information.

[2] Suchitwa Mission and Haritha Keralam Mission, Government of Kerala. Programme documents on Haritha Karma Sena.

[3] Kerala Solid Waste Management Project (KSWMP), World Bank-supported project documents.

Exam Revision Sheet | Case Study 7: Clean Kerala Company

30-SECOND ANSWER

Clean Kerala Company (CKCL) = **state company managing non-biodegradable and hazardous waste**; works with **Haritha Karma Sena** (women's door-to-door collectors, user fee) through MCFs and authorised recyclers. Strength: decentralised community model. Weakness: financing, fee collection, plastic prices.

KEY FACTS

- HKS started about 2017
- User fee paid to collectors
- MCFs aggregate and sort
- E-waste drives to authorised recyclers
- Plastic used in road tarring

KEY TERMS

HKS: Haritha Karma Sena, green army

MCF: material collection facility

LSGI: local self-government institution

RRF: resource recovery facility

NUMBERS & DATES TO REMEMBER

HKS about 2017 | Tens of thousands of workers

STRENGTHS / WHAT WORKED

- Community-based
- Women jobs
- Decentralised
- High awareness

LIMITS / CRITICISMS

- Fee issues
- Price volatility
- Low-value plastic

LIKELY EXAM QUESTIONS

1. Explain the role of CKCL and Haritha Karma Sena.
2. How can Kerala's model be replicated in other states?
3. Discuss its strengths and limitations.

Memory hook: "CKCL = Company + Karma Sena + MCF": three parts of the chain.

8. Pune's SWaCH Model

1. Executive Summary

SWaCH Seva Sahakari Sanstha Maryadit is a wholly owned cooperative of waste pickers and other urban poor in Pune, formed in 2008 in partnership with the Pune Municipal Corporation (PMC) and the waste-picker union KKPKP. Its members provide door-to-door collection of segregated waste, are paid mainly through user fees, and keep the recyclable material they recover. It is widely cited as a model that improves segregation, reduces municipal cost and provides dignified work. Its limitations are financial stability, dependence on municipal contracts and household compliance.

2. Background

Waste pickers in Pune had organised since 1993 through **Kagad Kach Patra Kashtakari Panchayat (KKPKP)**, the union of waste pickers and itinerant waste buyers, which gave them ID cards, health insurance and collective bargaining. The Municipal Solid Wastes (Handling) Rules and later SWM Rules 2016 required source segregation and door-to-door collection.

In 2008, KKPKP and PMC agreed to set up SWaCH to provide door-to-door service. PMC recognised the cooperative, provided ID cards and support, and allowed user fees to be collected directly by SWaCH.

3. Strategy / Initiative

- **Member-owned cooperative:** waste pickers are owners and service providers, not contractors' employees.
- **Door-to-door collection of segregated waste:** roughly 3,000 or more members serve several lakh households and commercial establishments (figures vary by year).
- **User-fee financing:** households and businesses pay a small monthly fee (typically in the tens of rupees); members keep the income and the value of recyclables.
- **Segregation and awareness:** members teach households, and SWaCH runs training and community campaigns.
- **Social protection:** ID cards, insurance, health and children's education support with KKPKP.
- **Partnership with PMC:** recognition, agreements and collaboration on decentralised composting and MRFs.

4. Impact & Results

Indicator	Result / data	Source
Established	2008; wholly owned cooperative of waste pickers	[1]
Members	About 3,000+ waste pickers and other workers	[1]
Households served	Several lakh households (around 9-10 lakh in some reports)	[1][2]
Waste diverted from landfill	Recycling reduces PMC transport and landfill load; independent studies show savings in crores of rupees annually	[2]
Worker outcomes	Higher and more stable incomes, ID cards, insurance	[1]
Recognition	Used as a model in policy discussion for integrating informal workers	[2]

SWaCH's membership and coverage vary yearly, and the contract with PMC has been renegotiated more than once; confirm the current numbers from the SWaCH website or annual report.

5. SWOT Analysis

STRENGTHS	WEAKNESSES
• Worker-owned, rights-based model • Strong segregation and door-to-door service • Trusted by households • Partnership with PMC	• Fee collection depends on households • Financial dependence on PMC support • Limited handling of low-value plastics • Physical strain and low pay for some members
OPPORTUNITIES	**THREATS**
• Expansion to e-waste and bulk generators • EPR funding for collection • Replication in other cities	• Privatisation or contractor takeover • Delays or changes in municipal contracts • Volatile scrap prices

6. Comparison with India

Aspect	Pune (SWaCH)	India
Model	Worker cooperative in partnership with ULB	Mostly contractors or municipal staff; pickers informal
Financing	User fee + sale of recyclables	ULB budgets, tipping fees, low user fees

Aspect	Pune (SWaCH)	India
Segregation	Household training by collectors	Weak in many cities
Worker rights	ID cards, insurance, union support	Rarely provided
Scale	City-wide in Pune	Few cities have comparable cooperatives

7. Recommendations

1. Give waste-picker cooperatives priority contracts, as allowed under SWM Rules 2016.
2. Ensure timely payments and stable municipal support.
3. Add e-waste, batteries and bulky waste collection to the service.
4. Link cooperatives to EPR funds and PROs.
5. Use apps and QR codes for household registration and fee payment.

8. Conclusion

SWaCH proves that integrating informal workers can improve service, dignity and recycling at the same time. Its lessons are directly relevant for India's EPR and e-waste chain, where informal collectors are still outside the formal system.

9. References

[1] SWaCH Seva Sahakari Sanstha Maryadit. Official website and annual reports.

[2] KKPKP and academic studies on SWaCH, including work by Chikarmane and Narayan on waste-picker integration in Pune.

[3] MoEFCC. Solid Waste Management Rules, 2016.

Exam Revision Sheet | Case Study 8: Pune SWaCH

30-SECOND ANSWER

SWaCH (2008) = **worker-owned cooperative of waste pickers** with PMC and KKPKP. Door-to-door collection of segregated waste; funded by **user fees + sale of recyclables**; gives ID cards, insurance, dignity. Risks: contract dependence and privatisation.

KEY FACTS

- KKPKP union: 1993
- SWaCH: 2008, cooperative
- User fee + recyclables income
- Members are owners, not employees
- SWM Rules 2016 recognise informal sector

KEY TERMS

SWaCH: Solid Waste Collection and Handling

KKPKP: waste-picker union

PMC: Pune Municipal Corporation

ULB: urban local body

NUMBERS & DATES TO REMEMBER

1993 KKPKP | 2008 SWaCH | About 3,000+ members

STRENGTHS / WHAT WORKED

- Rights-based
- Segregation
- Cost savings for city
- Replicable

LIMITS / CRITICISMS

- Fee dependence
- Contract risk
- Low-value plastics

LIKELY EXAM QUESTIONS

1. Describe the SWaCH model and its impact.
2. How does SWaCH integrate informal workers?
3. Can the model be used for e-waste collection?

Memory hook: "Own-Serve-Earn": pickers Own the coop, Serve households, Earn from fees and scrap.

9. Dharavi Recycling Ecosystem

1. Executive Summary

Dharavi, a 2.1 sq km settlement in the middle of Mumbai, hosts one of the world's densest informal recycling economies. Thousands of small units recycle plastic, metal, paper, cloth, leather and electronics collected from across Mumbai, employing very large numbers of workers and generating annual turnover often estimated near US$1 billion. It provides cheap, low-carbon material recovery, but with poor working conditions and health risks. The Dharavi Redevelopment Project raises the question of how to upgrade infrastructure without destroying this ecosystem.

2. Background

Dharavi began in the 19th century and grew as migrants settled on marshland and set up tanneries, pottery and later recycling. Estimates of its population range from about 6.5 lakh to over 10 lakh. It has about 13,000 single-room workshops (widely cited), many involved in recycling and small manufacturing.

Waste flows from across Mumbai via kabadiwalas and scrap traders. Plastic, especially, is sorted, washed, shredded and converted to granules for manufacturers; a large share of Mumbai's plastic recycling is estimated to happen here.

3. Strategy / Initiative

- **Multi-stage value chain:** collection, sorting by polymer and colour, washing, shredding, melting, pelletising and sale; parallel chains for aluminium, tin, paper, cloth and leather.
- **Specialised micro-units:** each unit does one step, with trust-based credit and quick trade.
- **Proximity and low cost:** short transport, cheap labour and space; high material recovery.
- **Informal finance and networks:** community-based credit, caste and regional networks, and traders linking Dharavi to national markets.
- **Redevelopment plan:** the Dharavi Redevelopment Project (Adani-led consortium selected in 2022) promises formal housing and infrastructure, with a proposed industrial or recycling component; the fate of existing units and workers is contested.

4. Impact & Results

Indicator	Result / data	Source
Area	About 2.1 sq km in central Mumbai	[1]
Population	Estimates about 6.5 lakh to 10 lakh+	[1]
Workshops	About 13,000 single-room units (widely cited)	[1][2]
Workers in recycling	Estimates from about 1 lakh up to 2.5 lakh	[2]
Annual turnover	Often estimated around US$1 billion for Dharavi's informal economy	[2]
Plastic share	A large share (some estimates 60-80%) of Mumbai's plastic recycling	[2]

Dharavi statistics are estimates from surveys and media reports. State them as approximate and cite the source and year.

5. SWOT Analysis

STRENGTHS	WEAKNESSES
• Very high recycling efficiency and skills • Low cost and low transport • Large employment • Strong trade networks	• Hazardous conditions, burns, chemical exposure • Unregulated and untaxed • Weak infrastructure and sanitation • Poor worker rights
OPPORTUNITIES	**THREATS**
• Upgrade with safe common facilities and training • Link to EPR and PROs • Cleaner technologies and finance	• Displacement due to redevelopment • Loss of workshop space and livelihoods • Volatile scrap prices and regulation

6. Comparison with India

Aspect	Dharavi	India
Recycling model	Dense informal clusters; manual	Formal MRFs and plants, mechanised
Efficiency	High recovery, low cost	Formal plants often under-utilised due to input shortage
Safety	Poor; open burning, chemicals	Regulated, PPE, emissions control

Aspect	Dharavi	India
Policy	Redevelopment issues, no legal recognition of units	SWM Rules 2016 recognise informal sector
Scale	Unique concentration in Mumbai	Similar clusters: Seelampur (Delhi), Moradabad

7. Recommendations

1. Provide in-situ or nearby industrial space for recycling units in the redevelopment plan.
2. Set up common effluent treatment and safe processing hubs.
3. Register workers and units; provide health insurance, skills and credit.
4. Link Dharavi recyclers to EPR/PRO funding for verified volumes.
5. Independent studies to map jobs, flows and environmental impact before redevelopment decisions.

8. Conclusion

Dharavi shows that informal recycling can achieve high recovery at low cost, but at high social and health cost. The best path is to formalise and upgrade, not to erase: giving the ecosystem space, safety and recognition within Mumbai's waste and EPR systems.

9. References

[1] Dharavi Redevelopment Project / Slum Rehabilitation Authority, Government of Maharashtra. Project documents and tender notices (2022-25).
[2] Academic and journalistic studies on Dharavi's economy (cite specific articles used, e.g. from BBC, Reuters or peer-reviewed journals).
[3] MoEFCC. Solid Waste Management Rules, 2016 (informal sector integration provisions).

Exam Revision Sheet | Case Study 9: Dharavi Recycling

30-SECOND ANSWER

Dharavi = **informal recycling cluster in Mumbai**: about 2.1 sq km, about 13,000 units, huge employment, about US$1 billion economy, large share of Mumbai's plastic recycling. **Efficient and cheap but hazardous.** Redevelopment threatens livelihoods; solution is formalise, protect and upgrade.

KEY FACTS

- Area about 2.1 sq km
- About 13,000 single-room workshops
- Plastic, metal, paper, cloth, leather recycling
- Redevelopment project (Adani-led, 2022)
- Kabadiwalas feed the chain

KEY TERMS

Informal sector: unregistered, low-regulation economy

Pelletising: converting plastic to granules

In-situ: at the same place

NUMBERS & DATES TO REMEMBER

2.1 sq km | 13,000 units | About US$1 bn | 2022 redevelopment award

STRENGTHS / WHAT WORKED

- High recovery
- Low cost
- Employment
- Skills

LIMITS / CRITICISMS

- Health hazards
- Untaxed
- Displacement risk

LIKELY EXAM QUESTIONS

1. Explain the Dharavi recycling value chain.
2. Should Dharavi be redeveloped or upgraded in place? Discuss.
3. Compare Dharavi with formal recycling.

Memory hook: "Small units, Big loop": tiny units, huge recovery.

10. Dell's Circular Economy Strategy

1. Executive Summary

Dell Technologies has set 2030 goals to make **100% of its product content from recycled or renewable material**, **100% of its packaging from recycled or renewable material**, and to **reuse or recycle an equivalent product for every product a customer buys**. It pursues these goals through design for repair and recycling, closed-loop plastics, take-back and reuse programmes, and supplier engagement. Dell's strategy demonstrates how a large electronics company can lead beyond compliance, but most claims are company-reported and need independent verification.

2. Background

Electronics have short lives, complex materials and high resource impact. Producers face EPR laws in many countries, including India's E-Waste Rules 2022, and customers and investors expect circularity. Dell's programme is described in its ESG reports under the "Progress Made Real" goals for 2030.

Dell has run consumer and business take-back services since the 2000s, including the Dell Reconnect partnership with Goodwill (from 2004), and reports recovering more than two billion pounds of used electronics since 2007.

3. Strategy / Initiative

- **Design:** use of recycled aluminium, plastics and other materials; modular and repairable designs; concepts such as Concept Luna for easy disassembly.
- **Closed-loop plastics:** using plastics from recycled electronics in new products (since around 2014) and ocean-bound plastics in packaging.
- **Packaging:** more moulded fibre and bamboo/wheat straw and less plastic.
- **Take-back and reuse:** recycling services, trade-in and refurbishment for business and consumers, via authorised recyclers, including in India under EPR.
- **Supply chain:** engagement with recyclers and suppliers for material traceability and responsible sourcing.
- **Goals and reporting:** 2030 targets and annual reporting to investors and stakeholders.

4. Impact & Results

Indicator	Result / data	Source
2030 goal: product content	100% recycled or renewable material	[1]
2030 goal: packaging	100% recycled or renewable material	[1]
2030 goal: take-back	Reuse or recycle an equivalent product for each one sold	[1]
Take-back volume	Over 2 billion lb (about 0.9 million t) recovered since 2007 (company-reported)	[1][2]
Closed-loop plastic	Recycled plastics from used electronics used in new products since about 2014	[2]
Independent check	Requires third-party assurance and comparison with peers	[3]

All Dell figures are company-reported. Use the latest ESG report for exact values and note that they are not independently verified.

5. SWOT Analysis

STRENGTHS

- Global scale and supplier influence
- Clear, time-bound goals
- Design for repair and recycling
- Established take-back network

WEAKNESSES

- Reliance on company-reported data
- Volumes small vs total sold
- Complex products limit material reuse
- Repairability varies by product line

OPPORTUNITIES

- Leading standards for circular products
- Growing refurbished and as-a-service markets
- Policy support through EPR and right-to-repair

THREATS

- Greenwashing criticism
- Recycled material supply and quality constraints
- Regulatory and competitor pressure

6. Comparison with India

Aspect	Dell	India
Approach	Producer-led global programme with goals	Mostly compliance-driven; few producers go beyond
Take-back	Consumer and enterprise, trade-in	EPR targets; low consumer take-back
Design	Recycled content, repairability	Limited eco-design requirement

Aspect	Dell	India
Reporting	Annual ESG reports	Portal reporting; few voluntary disclosures
Informal sector	Uses authorised recyclers	Informal chain dominant

7. Recommendations

1. Encourage Indian producers to publish recycled-content and take-back data.
2. Include eco-design and repairability criteria in EPR fees.
3. Support right-to-repair and spare-part availability.
4. Promote enterprise IT asset recovery with certified data destruction.
5. Encourage refurbished-device markets with quality standards.

8. Conclusion

Dell's strategy illustrates producer leadership in circularity, especially in design and take-back. For India, the lesson is to combine EPR compliance with eco-design, repair and transparent reporting, while ensuring independent verification of claims.

9. References

[1] Dell Technologies. ESG Report / Progress Made Real (latest edition), including 2030 goals.
[2] Dell Technologies. Circular economy and product recycling pages.
[3] MoEFCC. E-Waste (Management) Rules, 2022.

Exam Revision Sheet | Case Study 10: Dell Circular Economy

30-SECOND ANSWER

Dell's circular strategy: **2030 goals** of 100% recycled/renewable content and packaging, and reuse/recycle one product for every one sold. Tools: **design, closed-loop plastics, take-back, refurbishment.** Strong but mostly **company-reported**.

KEY FACTS

- Goals: content, packaging, take-back
- Dell Reconnect with Goodwill (2004)
- Closed-loop plastics since about 2014
- Over 2 billion lb recovered since 2007 (reported)
- Concept Luna: modular, repairable design

KEY TERMS

Circular economy: keep materials in use, design out waste

Closed-loop: recycled output returns to same product

Greenwashing: misleading environmental claims

NUMBERS & DATES TO REMEMBER

2030 goals | 2004 Reconnect | >2 billion lb since 2007

STRENGTHS / WHAT WORKED

- Clear goals
- Design focus
- Take-back network

LIMITS / CRITICISMS

- Self-reported data
- Small share of sales
- Greenwashing risk

LIKELY EXAM QUESTIONS

1. Explain Dell's circular economy strategy and its 2030 goals.
2. How does producer-led circularity differ from EPR compliance?
3. Critically evaluate Dell's claims.

Memory hook: "Design-Take back-Reuse": 3 steps in Dell's loop.